Station challenge cards (predict first)

One card per challenge station. A team reads the card, writes a prediction FIRST, then tests it with the slinky and records what happened. Keep the slinky flat on the floor and never let go while it is stretched. The reasons are on the instructor answer key so predictions stay real.

1 Count the round trips

Send one sharp pulse down the stretched slinky. Predict how many times it will travel down and back before it fades out, then send it and count.

My prediction: ______________________ What I measured: _______________

2 Tension change

One teammate pulls the slinky a little tighter (do not overstretch). Predict whether the pulse comes back faster or slower than before, then time it.

My prediction: ______________________ What I measured: _______________

3 Pinch a wall

A teammate pinches the middle of the slinky to make a fixed point. Predict what the pulse does when it reaches the pinch, then send one and watch.

My prediction: ______________________ What I measured: _______________

4 Measure the speed

Time 4 to 6 full round trips along your measured lane, then divide the total distance by the total time. Record the speed in metres per second.

My prediction: ______________________ What I measured: _______________

5 Two pulses meet

Two teammates each send a pulse from opposite ends at the same time. Predict what happens when the two pulses meet in the middle, then try it.

My prediction: ______________________ What I measured: _______________

6 Big push or small push

Send a small pulse, then a big one along the same lane. Predict whether the big push travels faster, slower, or the same speed, then time both.

My prediction: ______________________ What I measured: _______________

Instructor answer key (do not hand out)

Keep this page with the instructor and print it once. Reveal a station's reason only after each team has recorded its prediction.

#	WHAT HAPPENS	WHY
1	**Fades after several trips**	Friction and air drag remove energy each pass, so the count is a measurement, not a fixed number.
2	**Tighter is faster**	Wave speed rises with tension (it grows with the square root of tension), so a tighter slinky returns the pulse sooner.
3	**Reflects, flipped**	The pinch acts as a fixed end, so the pulse bounces back inverted, just like it does off the held far end.
4	**Distance over time**	Speed = total distance divided by total time. Timing several round trips and dividing cancels most of the human timing error.
5	**They pass through**	The two pulses cross by superposition: where they overlap the coils add for an instant, then each pulse continues on unchanged.
6	**Same speed**	Wave speed depends on the slinky's tension and mass per length, not on how big the pulse is; the big one just carries more energy.